

Killian Embler

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Seeking Summer 2027 engineering internship

EDUCATION

Embry-Riddle Aeronautical University

Bachelor of Science - Engineering Physics

Daytona Beach, FL

Fall 2024 - Present

- **Area of Concentration:** Spacecraft Instrumentation; **Cumulative GPA:** 4.00/4.00; **Expected Graduation:** Spring 2028
- **Minors:** Applied Mathematics, Computer & Electrical Engineering, Computer Science, Chinese Studies
- **Involvement:** Sigma Pi Sigma - National Honors Society for Physics and Astronomy; Society of Physics Students; Astrophysics Research Club

SKILLS

- **Computer Aided Design:** Inventor, Fusion 360, CATIA, Siemens NX, FEA, EasyEDA, KiCad, URDF Modeling
- **Manufacturing:** Drafting, Tolerances, CNC Lathes, Mills, Welding, 3D Printing (FDM), Laser Cutting, Hand and Power Tools
- **Electrical & Embedded:** Wiring, Soldering, PCB Design, Arduino, STM32, Raspberry Pi, Teensy, FPGA, Oscilloscope
- **Programming:** C/C++, Java, ROS/ROS2, ARM Assembly, Python, MATLAB, Verilog
- **Organizational:** Word, LaTeX, Excel, Agile Scrum, Experimental Design, Experimentation with Human Participants

EXPERIENCE

Embler Control Systems

Chief Technical Officer & Technical Co-Founder

Daytona Beach, FL

Sep 2025 - Present

- Co-founded a 12-person aerospace startup developing advanced flight computers for small to medium scale rocketry.
- Spearheaded research and development for the flight computer PCB, hardware implementation of testbeds, organization of launches for experimental validation and product iteration, and firmware for the computer during field operation.
- Presented in the Volusia County Pitch Competition, where Embler Control Systems won “Best Technical Startup.”
- Navigated the process of product development under FCC unintentional radiator constraints, ITAR standards for controlled model rocketry, and FAA safety regulations for test flights.

Robotics Intelligent Learning Lab (Mechanical Engineering Department)

Employed Undergraduate Researcher (Part-Time) / Project Lead

Daytona Beach, FL

Fall 2025 - Present

- Developing robot-human interactive control interfaces, with focus on assistive exoskeleton systems.
- Project Lead for both the hip isolated and full lower limb exoskeletons; submit budget requests and maintain weekly documentation.

Engineering Physics Propulsion Lab (EPPL)

Undergraduate Researcher

Daytona Beach, FL

Fall 2024 - Present

- Developing externally funded research projects with focus on the OpenMutt quadruped and OpenArm robotic arm.
- Design, manufacture, and collect test data on robotic platforms, present to and communicate with project supporters to secure funding.

PROJECTS

LECTER - Full Lower Limb Exoskeleton

Project Lead - Robotics Intelligent Learning Lab

- 14-DOF lower limb exoskeleton for human-robot control research. Contributed to the lab’s receipt of a \$523,000 grant from the NIH.
- Created URDF model, designed mechanical components including dual-linkage ankles for multi-axis actuation, redesigned wiring and power system to battery power with custom PCB and dual E-stop. Tuned impedance controller in ROS2 to assist squatting and sit-to-stand motions.
- First author of an extended abstract accepted to IROS 2026 (IEEE/RSJ International Conference on Intelligent Robots and Systems): When Muscles Think workshop, lead presenter of work at the conference.

SAMSARA - Flight Computer

Embler Control Systems

- Self-developed rocketry flight computer.
- 4-layer PCB built around a 32-bit ARM Cortex-M7 processor with an onboard IMU and altimeter, custom integrated DCDC buck converter, and multiple continuity sensing pyro lines, servo ports, and SPI sensor ports.
- Leading development of firmware for the system.

ALTAI - TVC Rocket

Embler Control Systems

- Thrust-vector-controlled rocket developed as an advanced flight computer testbed.
- Designed and manufactured internal rocket architecture to secure sensitive equipment during flight, deploy a recovery stage electronically without need for pyro lines or deployment charges, and stabilize the rocket without fins via a two-axis gimbal.
- Led development of stabilization control software using quaternion analysis that saw success across multiple low-altitude launches.

3D Printed Hip Isolated Exoskeleton

Project Lead - Robotics Intelligent Learning Lab

- 4-DOF hip exoskeleton as machine learning testbed for metabolic cost reduction research. Redesigned mechanical framework with FEA validation, added hip-width adjustability, produced URDF model, and designed the entire electrical architecture including a dedicated PCB.

OpenArm

Engineering Physics Propulsion Lab

- Open-source 5-DOF robotic arm designed to mount on the back of the OpenMutt quadruped. Led mechanical design and URDF creation, and worked on cycloidal gearbox assembly.
- Redesigned arm at reduced scale upon investor request to be implemented as payload for a Unitree Go2 quadruped. Operated under strict time constraints and investor requirements to perform static analysis of the rescaled arm, spec new actuators, design and manufacture the new arm, and implement a demo in ROS2 for the payload to function at a showcase.

AWARDS

- NASA Florida Space Grant Recipient - for Exoskeleton research under the Robotics Intelligent Learning Lab
- Embry-Riddle Diamond Eagle Scholarship; Paul B. Hunter Memorial Endowed Scholarship; Dean’s List - Fall 2024 until Present